Polarization of Reflection and Refraction Light

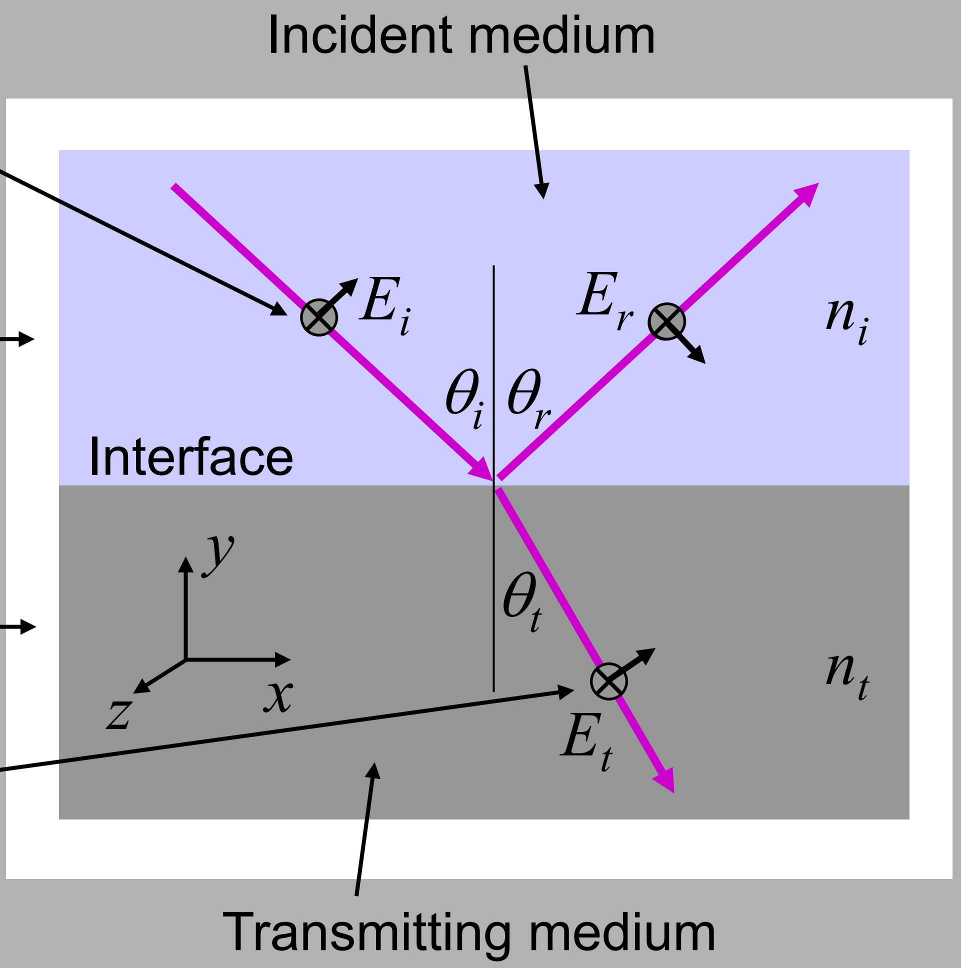
- Incident, transmitted, and reflected beams at interfaces
- The Fresnel Equations
- Critical Angle
- Brewster's Angle
- Total internal reflection

Definitions: Planes of Incidence and the Interface and the polarizations

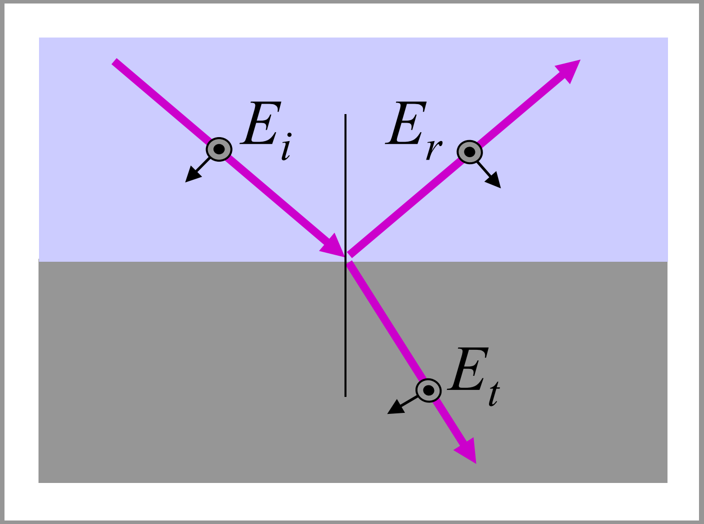
Perpendicular ("S") polarization sticks out of or into the plane of incidence.

Plane of incidence (here the xy plane) is the plane that contains the incident and reflected vectors.

Parallel ("P") polarization lies parallel to the plane of incidence.



Fresnel Equations



We would like to compute the fraction of a light wave reflected and transmitted by a flat interface between two media with different refractive indices.

$$r_\perp = E_{0r} / E_{0i}$$
$$t_\perp = E_{0t} / E_{0i}$$

for the perpendicular polarization

$$r_\parallel = E_{0r} / E_{0i}$$
$$t_\parallel = E_{0t} / E_{0i}$$

for the parallel polarization

where E_{0i}, E_{0r}, and E_{0t} are the field complex amplitudes.

Reflection & Transmission Coefficients for Perpendicularly Polarized Light

Solving for E_{0r} / E_{0i} yields the reflection coefficient :

$$\boxed{r_\perp = E_{0r} / E_{0i} = \left[n_i \cos(\theta_i) - n_t \cos(\theta_t)\right] / \left[n_i \cos(\theta_i) + n_t \cos(\theta_t)\right]}$$

Analogously, the transmission coefficient, E_{0t} / E_{0i}, is

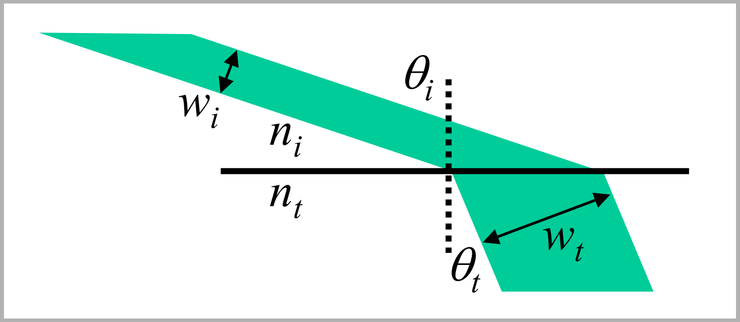
$$\boxed{t_\perp = E_{0t} / E_{0i} = 2n_i \cos(\theta_i) / \left[n_i \cos(\theta_i) + n_t \cos(\theta_t)\right]}$$

These equations are called the Fresnel Equations for perpendicularly polarized light.

Simpler expressions for $r_\perp$ and $t_\perp$

Recall the magnification at an interface, m:

$$m = \frac{w_t}{w_i} = \frac{\cos(\theta_t)}{\cos(\theta_i)}$$



Also let ρ be the ratio of the refractive indices, n_t / n_i.

Dividing numerator and denominator of r and t by $n_i \cos(\theta_i)$:

$$r_\perp = \left[n_i \cos(\theta_i) - n_t \cos(\theta_t)\right] / \left[n_i \cos(\theta_i) + n_t \cos(\theta_t)\right] = \left[1 - \rho m\right] / \left[1 + \rho m\right]$$

$$t_\perp = 2 n_i \cos(\theta_i) / \left[n_i \cos(\theta_i) + n_t \cos(\theta_t)\right] = 2 / \left[1 + \rho m\right]$$

$$\boxed{r_\perp = \frac{1 - \rho m}{1 + \rho m}} \qquad \boxed{t_\perp = \frac{2}{1 + \rho m}}$$

Reflection & Transmission Coefficients for Parallel (P) Polarized Light

Solving for E_{0r} / E_{0i} yields the reflection coefficient, $r_{||}$:

$$\boxed{r_{||} = E_{0r} / E_{0i} = \left[n_i \cos(\theta_t) - n_t \cos(\theta_i)\right] / \left[n_i \cos(\theta_t) + n_t \cos(\theta_i)\right]}$$

Analogously, the transmission coefficient, $t_{||} = E_{0t} / E_{0i}$, is

$$\boxed{t_{||} = E_{0t} / E_{0i} = 2n_i \cos(\theta_i) / \left[n_i \cos(\theta_t) + n_t \cos(\theta_i)\right]}$$

These equations are called the Fresnel Equations for **parallel** polarized light.

Simpler expressions for $r_\parallel$ and $t_\parallel$

$$r_\parallel = E_{0r} / E_{0i} = \left[n_i \cos(\theta_t) - n_t \cos(\theta_i) \right] / \left[n_i \cos(\theta_t) + n_t \cos(\theta_i) \right]$$

$$t_\parallel = E_{0t} / E_{0i} = 2 n_i \cos(\theta_i) / \left[n_i \cos(\theta_t) + n_t \cos(\theta_i) \right]$$

Again, use the magnification, m, and the refractive-index ratio, ρ .

And again dividing numerator and denominator of r and t by $n_i \cos(\theta_i)$:

$$r_\parallel = \left[m - \rho \right] / \left[m + \rho \right]$$

$$t_\parallel = 2 / \left[m + \rho \right]$$

$$\boxed{r_\parallel = \frac{m - \rho}{m + \rho}} \qquad \boxed{t_\parallel = \frac{2}{m + \rho}}$$

Reflection Coefficients for an Air-to-Glass Interface

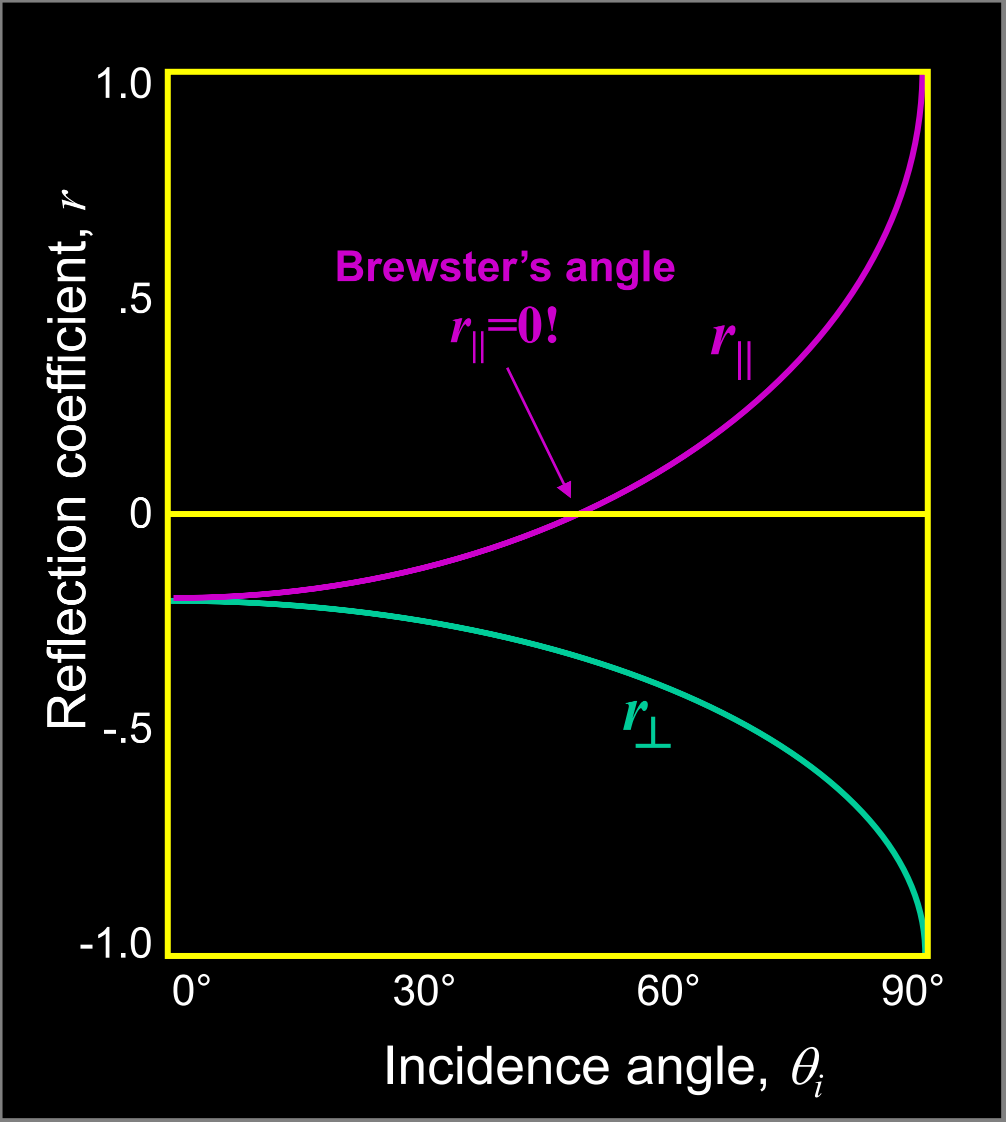
$$n_{air} \approx 1 \; < \; n_{glass} \approx 1.5$$

Note that:

Total reflection at θ = 90° for both polarizations

Zero reflection for parallel polarization at **Brewster's angle** (56.3° for these values of n_i and n_t).

$$\theta_B \equiv \arctan(n_t / n_i)$$



Reflection Coefficients for a Glass-to-Air Interface

$$n_{glass} \approx 1.5 \;>\; n_{air} \approx 1$$

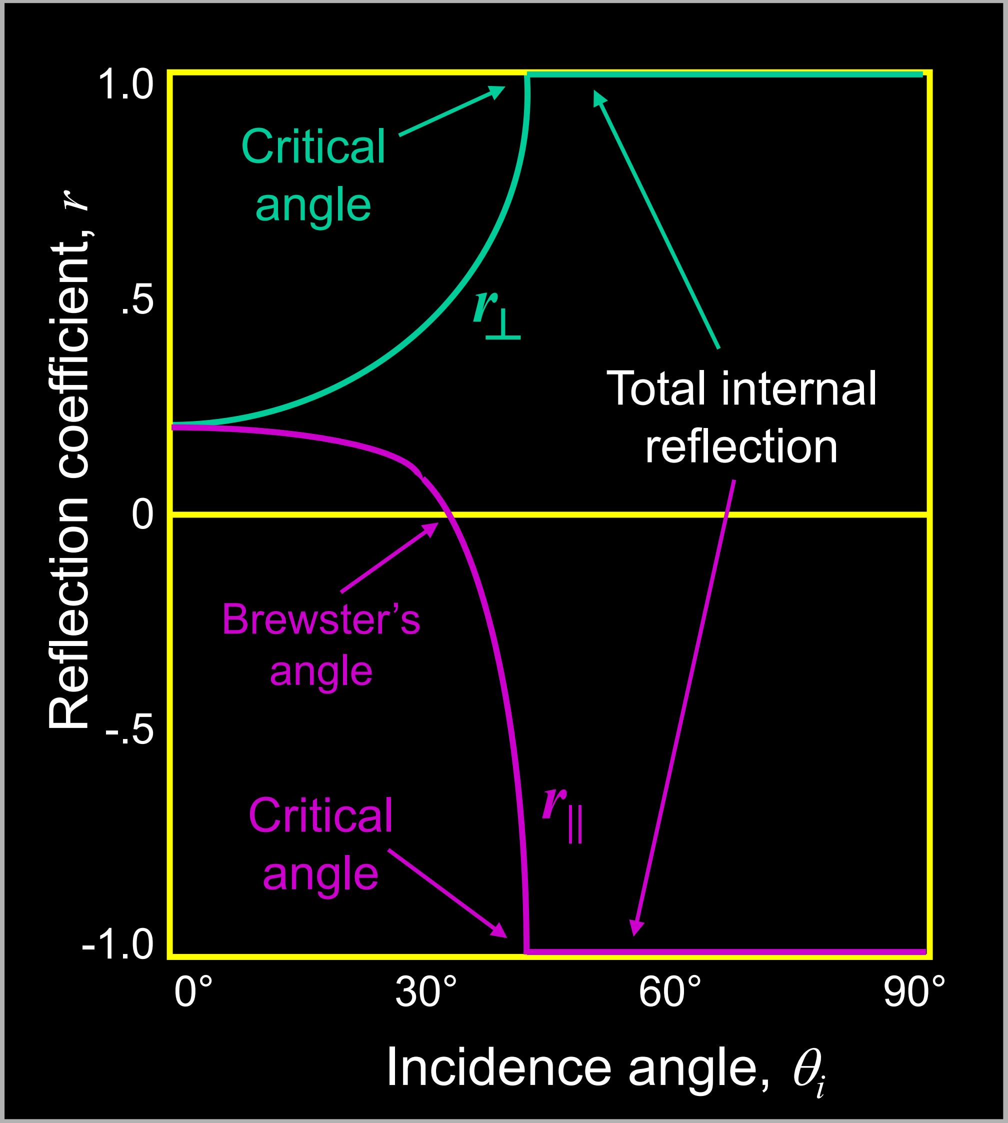
Note that:

Total internal reflection above the **critical angle**

$$\boxed{\theta_{crit} \equiv \arcsin(n_t / n_i)}$$

(The sine in Snell's Law can't be > 1!):

$$\sin(\theta_{crit}) = n_t / n_i \sin(90°)$$

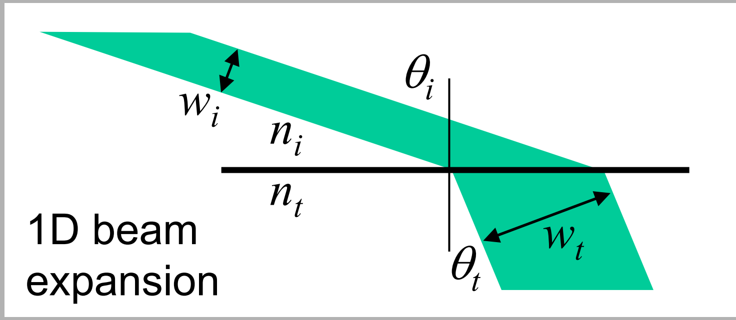


Transmittance (T)

$$I = \left(n \frac{\varepsilon_0 c_0}{2} \right) |E_0|^2$$

$T \equiv$ Transmitted Power / Incident Power $= \dfrac{I_t A_t}{I_i A_i}$ $\quad$ A = Area

Compute the ratio of the beam areas:



$$\frac{A_t}{A_i} = \frac{w_t}{w_i} = \frac{\cos(\theta_t)}{\cos(\theta_i)} = m$$

The beam expands in one dimension on refraction.

$$\frac{|E_{0t}|^2}{|E_{0i}|^2} = t^2$$

$$T = \frac{I_t A_t}{I_i A_i} = \frac{\left(n_t \frac{\varepsilon_0 c_0}{2} \right) |E_{0t}|^2}{\left(n_i \frac{\varepsilon_0 c_0}{2} \right) |E_{0i}|^2} \left[\frac{w_t}{w_i} \right] = \frac{n_t |E_{0t}|^2 w_t}{n_i |E_{0i}|^2 w_i} = \frac{n_t}{n_i} t^2 \frac{\cos(\theta_t)}{\cos(\theta_i)}$$

$$\Rightarrow \boxed{T = \left[\frac{(n_t \cos(\theta_t))}{(n_i \cos(\theta_i))} \right] t^2 = \rho\, m\, t^2}$$

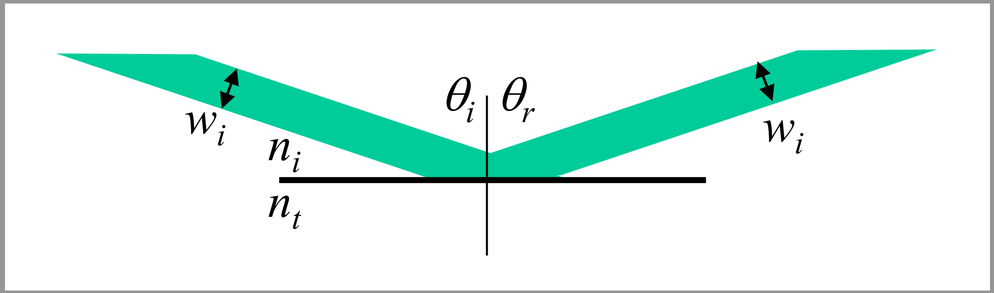
The Transmittance is also called the Transmissivity.

Reflectance (*R*)

$$I = \left(n \frac{\varepsilon_0 c_0}{2} \right) \left| E_0 \right|^2$$

$$R \equiv \text{Reflected Power / Incident Power} = \frac{I_r A_r}{I_i A_i}$$

A = Area



Because the angle of incidence = the angle of reflection, the beam area doesn't change on reflection.

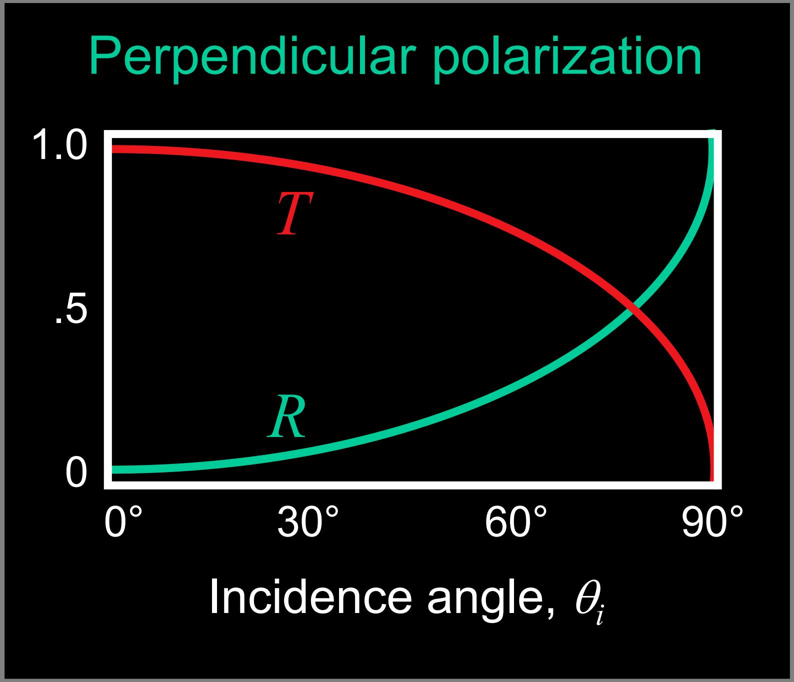
Also, n is the same for both incident and reflected beams.

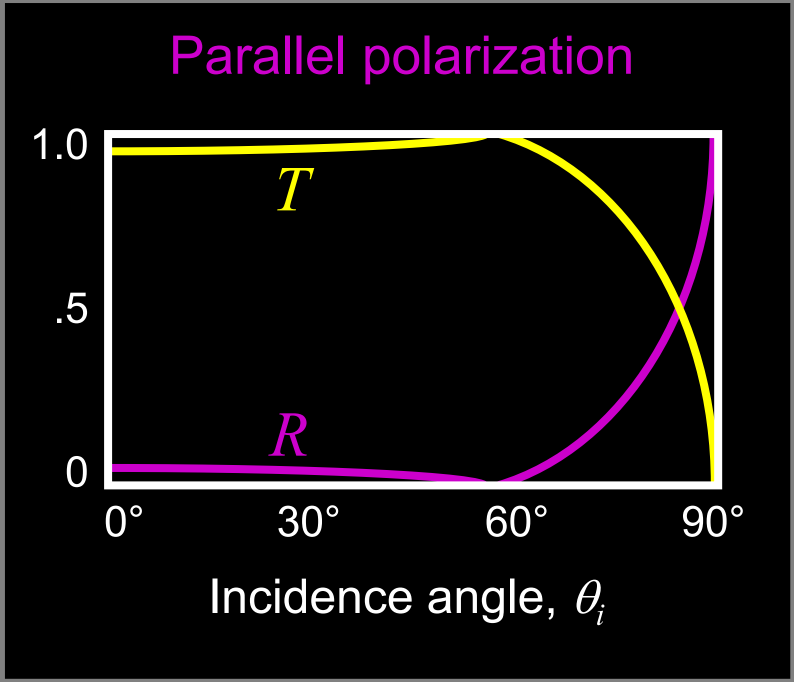
So:

$$\boxed{R = r^2}$$

The Reflectance is also called the Reflectivity.

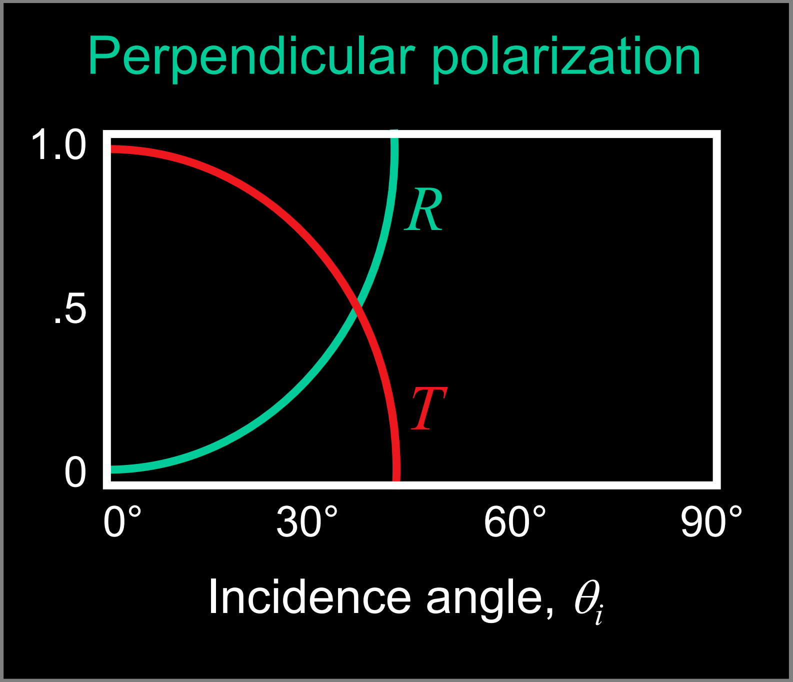
Reflectance and Transmittance for an Air-to-Glass Interface

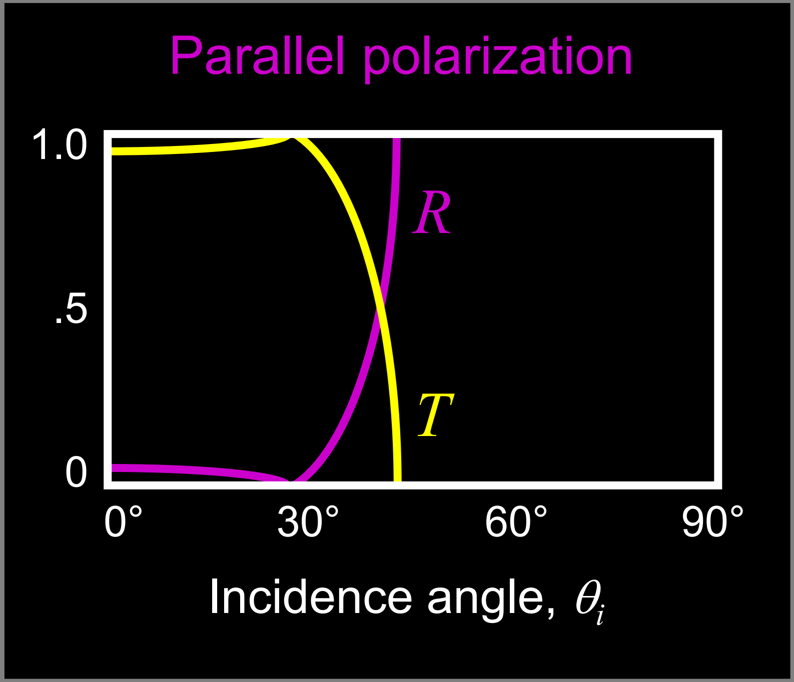




Note that $R + T = 1$

Reflectance and Transmittance for a Glass-to-Air Interface





Note that $R + T = 1$

Reflection at normal incidence

When $\theta_i = 0$,

$$R = \left(\frac{n_t - n_i}{n_t + n_i}\right)^2$$

and

$$T = \frac{4 n_t n_i}{\left(n_t + n_i\right)^2}$$

For an air-glass interface ($n_i = 1$ and $n_t = 1.5$),

R = 4% and T = 96%

The values are the same, whichever direction the light travels, from air to glass or from glass to air.

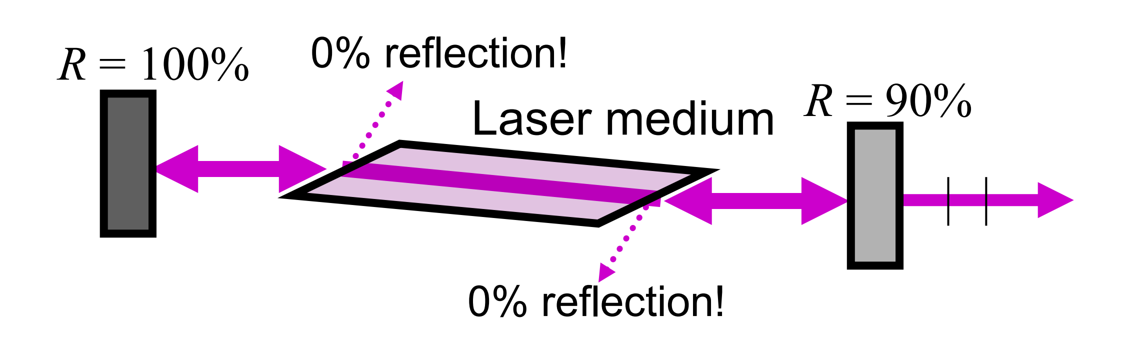
The 4% has big implications for photography lenses.

Practical Applications of Fresnel's Equations

Windows look like mirrors at night (when you're in the brightly lit room)

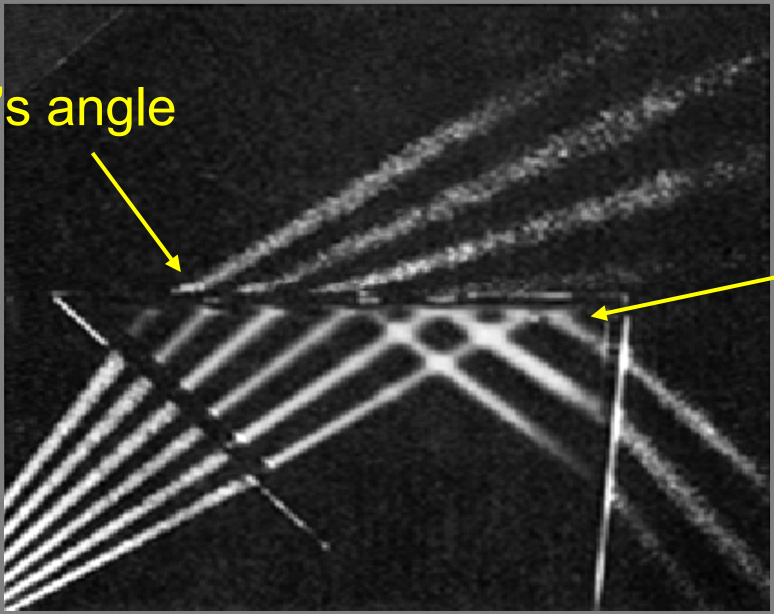
One-way mirrors (used by police to interrogate bad guys) are just partial reflectors (actually, aluminum-coated).

Lasers use Brewster's angle components to avoid reflective losses:



Total Internal Reflection occurs when $\sin(\theta_t) > 1$, and no transmitted beam can occur.

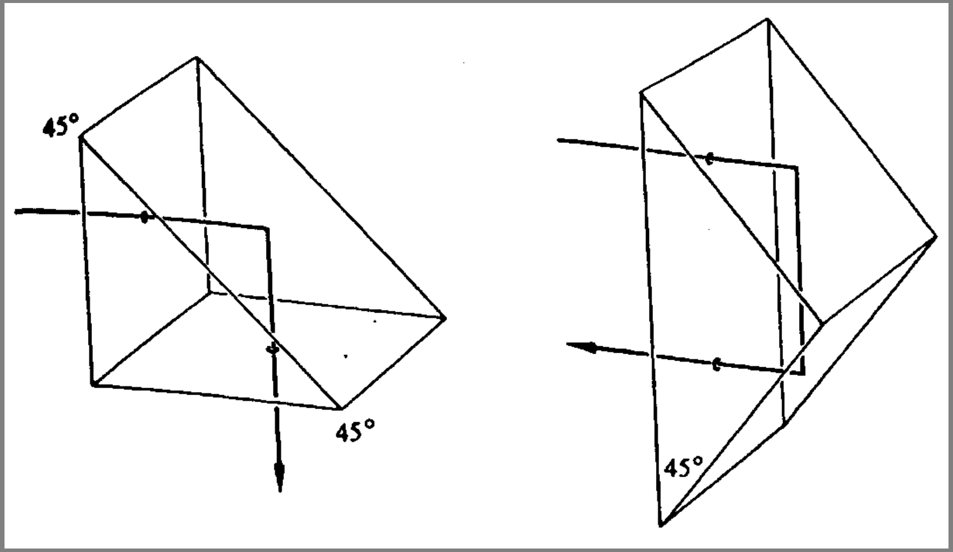
Note that the irradiance of the transmitted beam goes to zero (i.e., TIR occurs) as it grazes the surface.



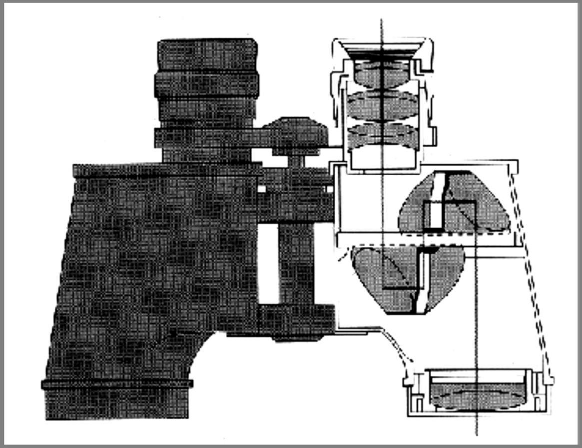
Total internal reflection is 100% efficient, that is, all the light is reflected.

Applications of Total Internal Reflection

Beam steerers

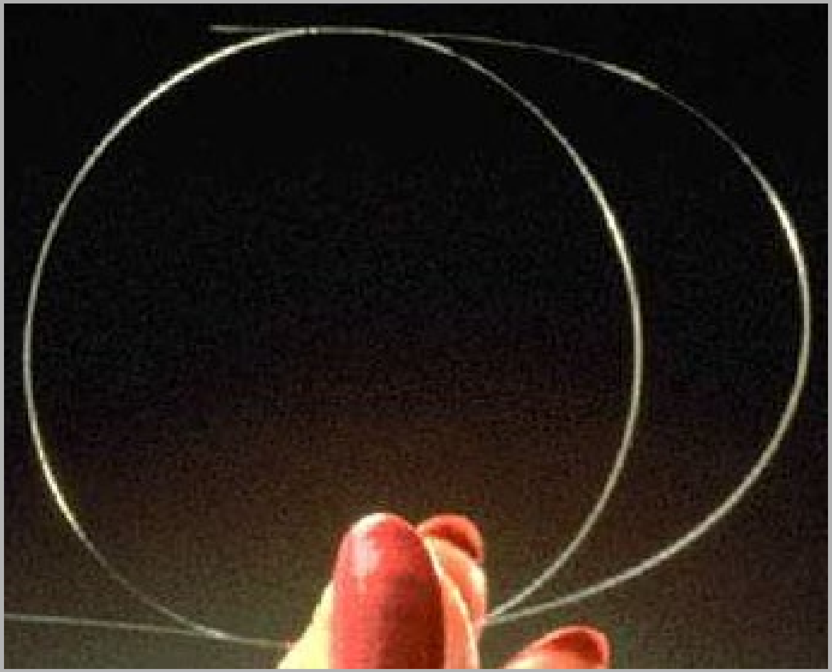


Beam steerers used to compress the path inside binoculars



Fiber Optics

Optical fibers use TIR to transmit light long distances.

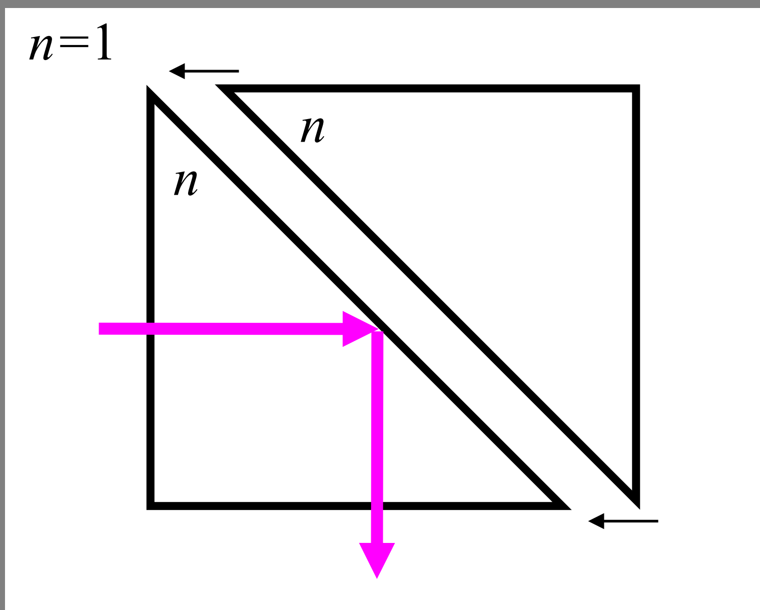


They play an ever-increasing role in our lives!

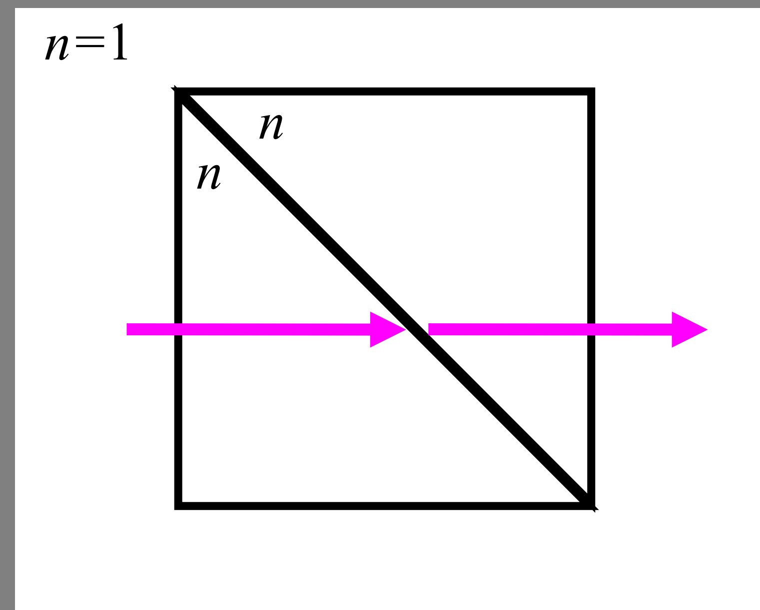
Frustrated Total Internal Reflection

By placing another surface in contact with a totally internally reflecting one, total internal reflection can be **frustrated**.

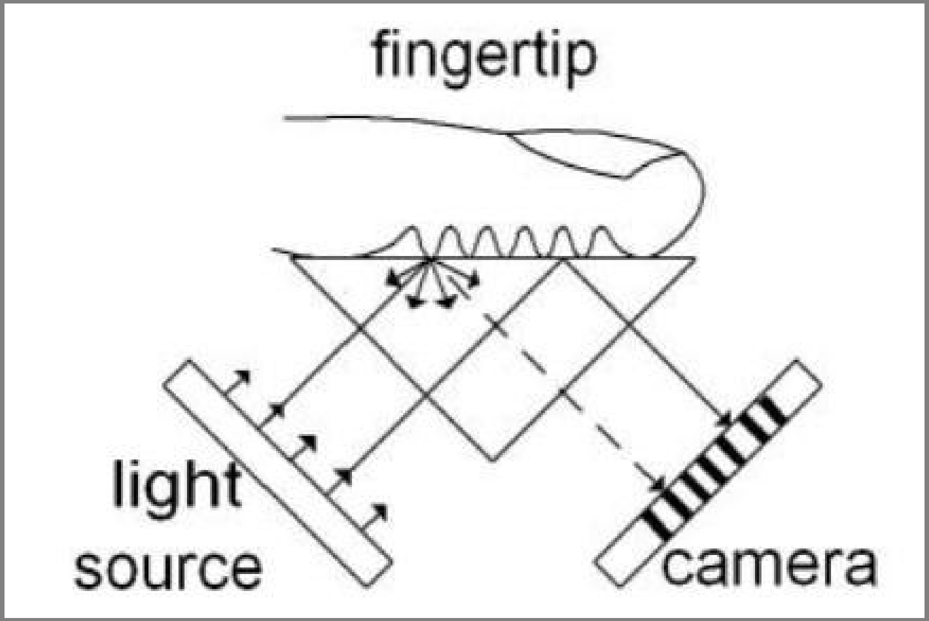
Total internal reflection

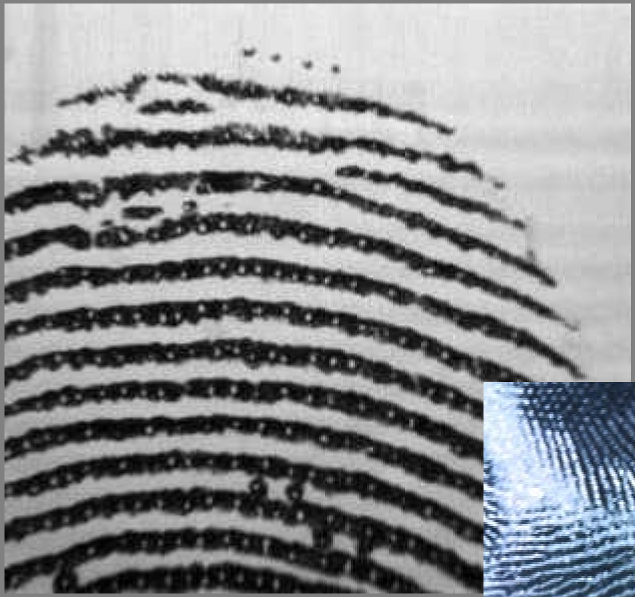


Frustrated total internal reflection



FTIR and fingerprinting





See TIR from a fingerprint valley and FTIR from a ridge.